Statistical methods for Machine Learning: project report

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1 Project Links

`https://github.com/Valementajat/ML_Stability_Regularization`

Google Colab Notebook:
`https://colab.research.google.com/github/Valementajat/ML_Stability_Regularization/blob/master/regularizationStability.ipynb`

2 Introduction

This report presents my work for Assignment 3 - Regularization and Stability. I will go through what I have achieved and what steps I have taken

The goal was to make an empirical study about the relationship between:

- regularization strength
- algorithmic stability
- generalization performance

For this goal, a set of requirements, or tasks was implemented in the project paper:

- Train models with varying regularization
  - Ridge regression
  - Unregularized least squares
- Measure:
  - Training error
  - Test error
- Estimate stability
  - For each training point

    - Remove (or replace) the point
    - Retrain the model
- Measure the change in predictions on a fixed test set
    - Define a stability metric, for example
        - Average change in prediction
        - Average change in loss
- Plot
    - Stability vs regularization strength
    - Test error vs regularization strength

Two datasets were used: the diabetes regression dataset from scikit-learn and a generated synthetic regression dataset. The final version of the project was last run on 23 April 2026. As a result, the empirical study was a success and evaluation metrics of relationship from before mentioned points have been made. The final version of the project is easily accessible from the link section at the start of the report.

3 Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

4 Datasets and Pre-processing

Two types of datasets were used in the project. A diabetes regression dataset imported from scikit and a synthetic dataset created with set noise.

To ensure that test results are replicable, for synthetic dataset, as we're using numpy library to generate randomness, we set a seed which ensures that we're generating the same dataset every time. As well when splitting the dataset into train, validation and test set, we're using scikit's built in `train_test_split` method and set a seed to get replicable result. Every other part follows a steady algorithm so no deviation is possible.

The data was split into training, validation, and test sets. First, 20% of the data was reserved as a test set. Then, 25% of the remaining data was used as a validation set. This produced an approximate 60/20/20 split for training, validation, and testing.

4.1 Synthetic dataset

The synthetic dataset was designed to make regularization useful. It contains more features than the real dataset and includes correlated features. In particular, the second and third features were constructed to be almost linear combinations of the first feature. This creates correlation, where several features contain almost the same information, which is something that Ridge regularization is expected to pick up, because it penalizes large weights and can reduce sensitivity to small changes in the training data.

5 Model and Regularization

This section describes the linear regression model used in the experiments and how ridge regularization was added to it.

The implemented model is a linear regression model of the form

$$\hat{y} = Xw,$$

where X is the feature matrix and w is the vector weights. A bias column of ones is added to the feature matrix so that the model can also learn an intercept term, A bias column of ones is added to the feature matrix:

$$X_{\text{bias}} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1d} \\ 1 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}.$$

The model prediction is then

$$\hat{y} = X_{\text{bias}} w.$$

For the unregularized model, the objective minimized is the mean squared error:

$$\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2.$$

For ridge regression, an L_2 penalty is added to the objective:

$$\frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 + \alpha \sum_{j=1}^{d} w_j^2.$$

The intercept term was not regularized. This was done by setting the regularization gradient of the first weight to zero.

Both models were trained using the same **ridge()** function but with different regularization variable alpha, where a **non** regularized model was achieved using $\alpha = 0$

5.1 Model

Due to the project's constraints, the model was trained using a custom gradient descent implementation rather than using scikit-learn's built-in ridge regression model. Scikit-learn was only used for loading the diabetes dataset and for splitting the data into training, validation, and test sets.

The model training process is the following:

1. The dataset is split into training, validation, and test sets
2. A bias column is added to the feature matrix so that the model can learn an intercept term.
3. For each value of the regularization variable α, a regression model is trained using gradient descent.
4. During training, the model minimizes the following objective:

   $$\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 + \alpha\sum_{j=1}^{d} w_j^2$$

   where the first term is the mean squared error and the second term is the ridge regularization penalty. By using an **alpha = 0**, we're effectively creating an unregularized least squares model
5. Calculate the gradient using two helper functions: `getFeatureBlame` for the mean squared error part and `getRidgeGradient` for the regularization part.

   The gradient combines two parts. First, the normal mean squared error gradient is calculated:

   $$\frac{2}{m}X^T(\hat{y} - y)$$

   Then, the ridge regularization gradient is added:

   $$2\alpha w$$

   The full gradient is therefore:

   $$\nabla = \frac{2}{m}X^T(\hat{y} - y) + 2\alpha w$$

   The bias term is not regularized. In the implementation, this is done by setting the first value of the regularization gradient to zero:

   ```
   reg[0] = 0
   ```
6. Update weights based on the gradient, using the learning rate.
7. Repeat the amount of set epochs.

8. For each α, different learning rates are tested, and the model with the lowest validation error is selected.

Thus we're left with a trained model, as in the weights.

6 Experimental Methodology

Algorithmic stability was estimated using the recommended leave-one-out retraining procedure. For each value of the regularization parameter α, a base model was first trained on the full training set. Then, each training example was removed one at a time, and the model was retrained using the same value of α. The retrained model was evaluated on the same fixed test set.

Two stability metrics were computed. The first metric measured the average absolute change in predictions:

$$\text{Stability}_{pred} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{m}\sum_{j=1}^{m}|\hat{y}_j^{base} - \hat{y}_j^i|,$$

where $\hat{y}_j^{base}$ is the prediction of the model trained on the full training set, and $\hat{y}_j^i$ is the prediction after removing the i-th training point.

The second metric measured the average absolute change in test MSE:

$$\text{Stability}_{loss} = \frac{1}{n}\sum_{i=1}^{n}\left|\text{MSE}_{test}^{base} - \text{MSE}_{test}^{i}\right|.$$

7 Limitations

There are several limitations in this empirical study. First, the ridge model was trained using gradient descent, rather than the closed-form ridge solution. Therefore, the results depend on the learning rate and number of epochs. To reduce this issue, several learning rates were tested and the best one was selected using the validation set.

Second, stability estimation was computationally expensive because the model was retrained once for each training example, for each value of α and multiple learning rates. For this reason, the experiment was performed on relatively small datasets.

Third, the prediction-change stability metric and the loss-change stability metric measure different aspects of stability. In the real dataset, these two metrics did not always move in the same direction. Therefore, stability should not be interpreted as a single absolute value, but rather as an empirical estimate depending on the chosen metric.

Finally, only two datasets were used. The synthetic dataset was designed to highlight the usefulness of ridge regularization, while the diabetes dataset represents a small real-world regression problem. More datasets would be needed to make broader conclusions.

8 Results

For the real diabetes dataset, increasing regularization generally increased both training and test error. The lowest test error score was obtained with $\alpha = 0$, prompting that the unregularized model performed best in terms of prediction accuracy on this dataset.

The loss-based stability metric decreased from approximately 4.16 at $\alpha = 0$ to approximately 1.82 at $\alpha = 0.1$, suggesting that moderate regularization made the model less sensitive in terms of test loss. However, the prediction-change stability metric did not decrease. Instead, it slightly increased with α.

This suggests that the diabetes dataset does not demonstrate the expected relationship as clearly as the synthetic dataset. A likely reason is that the diabetes dataset is relatively small and low-dimensional with only 10 features + bias. Therefore, the unregularized model was already reasonably stable, and stronger regularization mainly caused underfitting.

8.1 Synthetic Dataset

The synthetic dataset showed the expected relationship between regularization, stability, and generalization more clearly. With $\alpha = 0$, the model achieved a low training MSE but a much higher test MSE, suggesting overfitting.

As α increased, test error initially decreased. The best test MSE was obtained with $\alpha = 1$. At the same time, the average prediction change decreased, showing that the model became more stable.

For larger values of α, stability continued to improve, but test error started to increase again. This indicates underfitting: the model becomes less sensitive to changes in the training set, but it also loses predictive accuracy.

α	Train MSE	Test MSE	Stability Pred.	Stability Loss
0	0.713	15.529	0.314	0.616
0.01	0.737	15.177	0.308	0.597
0.1	1.011	12.798	0.258	0.464
1	3.909	9.532	0.117	0.194
3	7.878	11.850	0.074	0.145
5	10.531	13.021	0.051	0.095
10	13.915	16.086	0.053	0.123

Table 1: Synthetic dataset results.

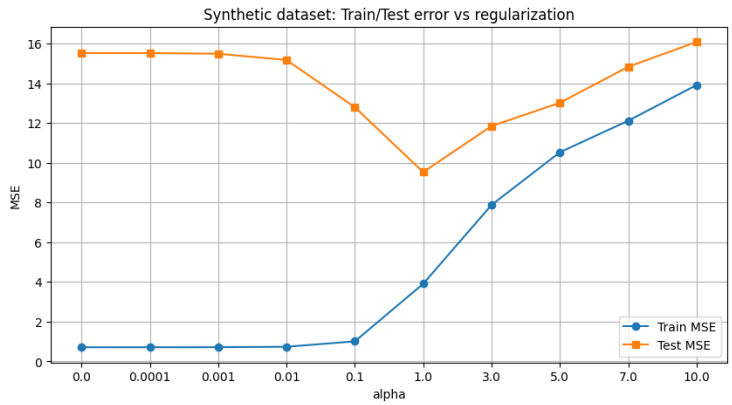

Figure 1: Training and test MSE as a function of α on the synthetic dataset. Moderate regularization gives the lowest test error.

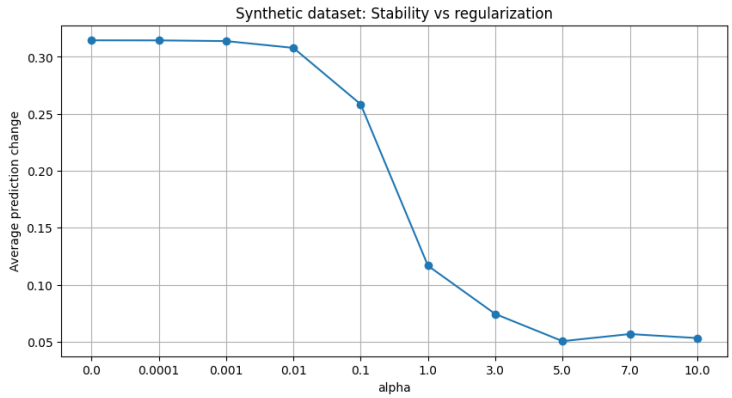

Figure 2: Average prediction change as a function of α on the synthetic dataset. Stability generally improves as regularization increases.

8.2 Real Diabetes Dataset

For the real diabetes dataset, increasing regularization generally increased both training and test error. The lowest test error was obtained with $\alpha = 0$, meaning that the unregularized model performed best in terms of prediction accuracy on this dataset.

The loss-based stability metric decreased from approximately 4.16 at $\alpha = 0$ to approximately 1.82 at $\alpha = 0.1$, suggesting that moderate regularization made the model less sensitive in terms of test loss. However, the prediction-change stability metric did not decrease. Instead, it slightly increased as α increased.

From this we can determine that the diabetes dataset does not demonstrate the expected relationship as clearly as the synthetic dataset. A likely reason is that the diabetes dataset is relatively small and low-dimensional, with only 10 original features. Therefore, the unregularized model was already reasonably stable, and stronger regularization mainly caused underfitting.

α	Train MSE	Test MSE	Stability Pred.	Stability Loss
0	4699.37	3929.88	0.226	4.165
0.001	4732.53	3958.80	0.226	4.127
0.01	4995.49	4190.81	0.229	3.724
0.1	5877.53	4995.24	0.246	1.824
1	6179.35	5278.99	0.253	1.952
10	6212.95	5310.85	0.253	2.021

Table 2: Real diabetes dataset results.

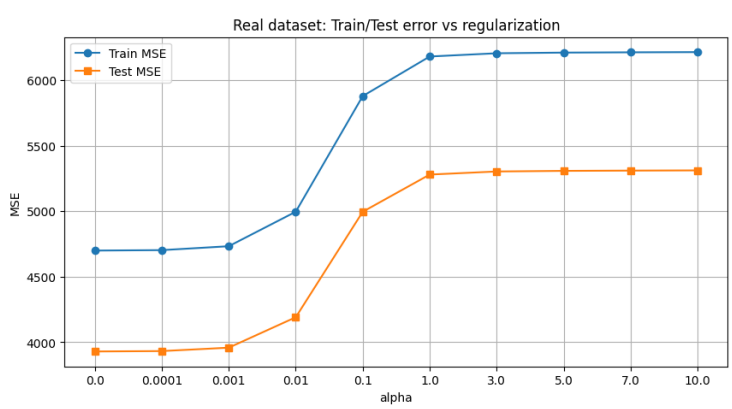

Figure 3: Training and test MSE as a function of α on the real diabetes dataset. The unregularized model gives the lowest test error.

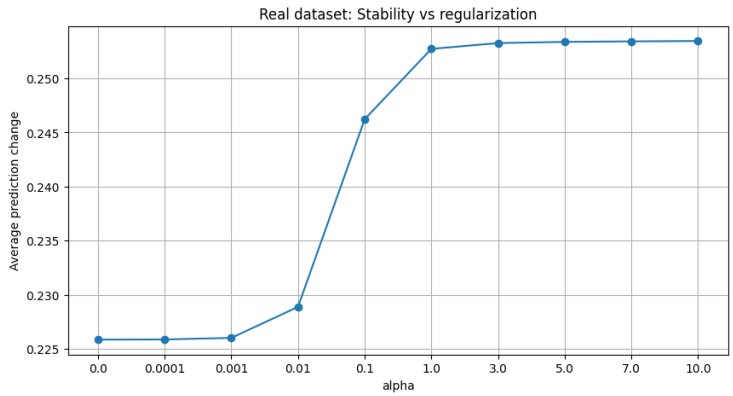

Figure 4: Average prediction change as a function of α on the real diabetes dataset. The trend differs from the synthetic dataset.

9 Conclusion

Generally speaking, the experiments show a relationship between regularization, algorithmic stability, and generalization performance. On the synthetic dataset, increasing α reduced the average prediction change, meaning that the model became more stable. Moderate regularization also improved test performance, with the best test MSE obtained at $\alpha = 1$.

However, the results also show that stronger regularization is not always better. On the synthetic dataset, very large values of α increased test error, which indicates underfitting. On the real diabetes dataset, the unregularized model achieved the lowest test error, and stronger regularization mostly made performance worse.

Therefore, the empirical results support the idea that regularization can improve stability and generalization, especially in noisy or high-dimensional settings. However, the best regularization strength depends on the dataset, and excessive regularization can reduce predictive accuracy.